CS5013 Mid-demo Report

Exchange Student Guide & Community Platform

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Repository: https://github.com/rikire/Exchange-Student-Guide-Platform. As in the design document, every figure below names the file or command it comes from, so it can be checked against the repository rather than taken on trust.

What the rubric asks	Section	Evidence in the repository
Core workflow on a real input	§1	`docs/course/mid-demo.md`
Both members contributing across weeks	§2	`scripts/contribution.sh`, `docs/team/`
An honest gap list	§3	`docs/gap-list.md`, `docs/tech-debt.md`
Explaining any part of the code	§4	`docs/ai/journal/`, `docs/verification/`

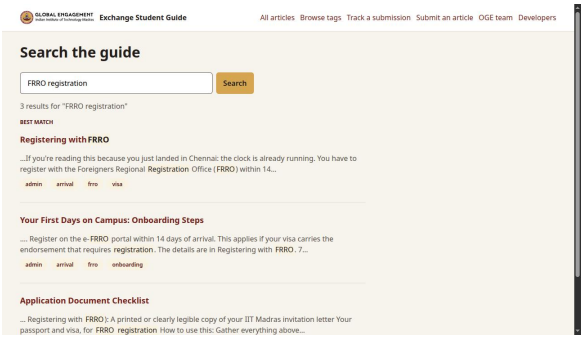
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1 The core workflow runs end to end on a real input

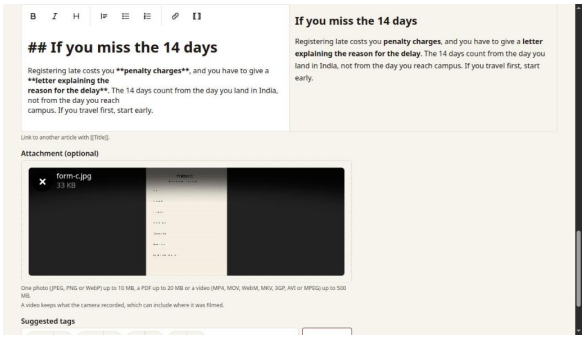
The scenario of the proposal: a student searches for "FRRO registration", reads the article, follows a wiki link, proposes an edit with a photo of a paper form; the OGE moderator approves it, and the change is live. The real input is the guide's seed: 38 articles written from OGE's International Student Handbook 2026, each citing its pages (`app/src/main/resources/data/seed/`). *Registering with FRRO* is one of them.

#	Who	Step and what is seen	Requirement, test
1	Student	The landing page: search box, pinned articles	FR-009, `LandingControllerTest`
2	Student	"FRRO registration" returns *Registering with FRRO* first	FR-007, `SearchFlowTest`
3	Student	The article, rendered from Markdown, with its tags	FR-001, `ArticleControllerTest`
4	Student	The wiki link *your student visa* opens *Applying for Your Student Visa*	FR-002, `WikiLinkRendererTest`
5	Student	*Propose an edit*: a sentence and a photo; a submission number; the article unchanged	FR-011, `SubmissionFlowTest`
6	Moderator	Signs in; the edit at the top of the queue	FR-014, `ModerationFlowTest`
7	Moderator	The change as a diff of the text as read, and the photo	FR-015, FR-029
8	Moderator	*Approve & publish*: the sentence and the photo are live, and found by search	FR-017, FR-032

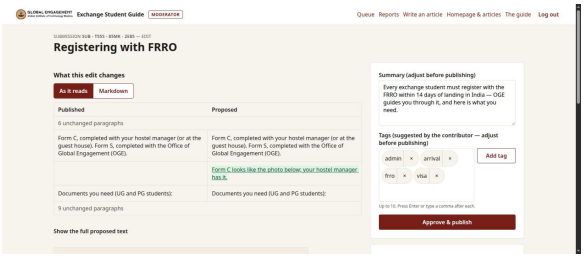
Run on 9 October on a stand reset from scratch (`docker compose down -v`, then `up --build`: 38 articles imported, the search index built at start), in Chrome; the edit was `SUB-T55S-85MK-2E85`, the photo a drawing of Form C.



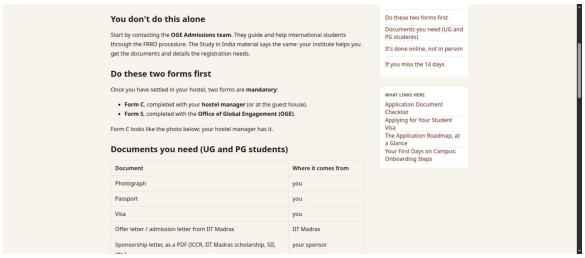
2. "FRRO registration", the article first



5. The edit: the photo previewed



7. The moderator's diff of the text as read



8. The sentence live in the article

The rehearsal found what the tests could not

On 4–5 October the scenario was run on the Docker Compose stand, with PostgreSQL and the volumes an earlier version had filled, rather than on the in-memory database the tests use (`docs/roadmap/03-main-flow.md`). Four faults surfaced that none of the 635 application tests could see: two lived in what an older image had left on disk, one in a page no test asked for, and one in a mouse click:

- The stand did not start: since the image runs as an unprivileged user, the volumes written by the earlier root image were read-only to it. Recorded as `DEBT-026`, with the one-off fix.
- The approval answered `500`: the search index on disk predated a change to its fields, and was kept because it held documents. Since 5 October it is rebuilt at every start, a reversal recorded as an amendment to ADR-0004.
- That `500` was Tomcat's bare page: no generic error page existed. Added, with a test.
- After a tag was added with Enter, its list of suggestions stayed open over *Submit for review*, so a click there added a tag and sent nothing. Fixed, with the browser test that would have caught it.

After the fixes the scenario ran end to end on the stand. The demo itself is given later; the stand is reset before it (`docker compose down -v`).

2 Both members contribute, across weeks

Authored commits per ISO week, from `sh scripts/contribution.sh`. Journal entries written by the `Stop` hook are counted apart and never added in.

	W35	W36	W37	W38	W39	W40	W41 (to 11 Oct)
Mikhail	1	36	34	–	63	34	28
Abdirakhim	–	8	9	–	39	174	25

Two things this table shows that we do not hide. Week 38 (14–20 September) has no commits from either of us; the weekly log for it is still to be agreed (`docs/team/weekly-log/`). And the split is uneven by area: `docs/team/ownership.md`, generated from the same history, gives Abdirakhim more commits in twelve of the thirteen application slices, while Mikhail built the process layer (`tools/`, 132 commits: the prompt journal, the guards, the generators of the gap list, traceability and ownership) and, in the application, the contributor's forms of fix 3.6.

3 The gap list is honest because nobody writes it

`docs/gap-list.md` is generated by `java -jar tools/target/ai-tools.jar gaps` from the requirements' status, the debt register and the traceability chain, and the pre-push hook refuses a push while it is out of date. It cannot be trimmed before a demo without changing what it is generated from.

- **Requirements:** 42 of 43 functional and non-functional requirements done; FR-028 (completing a wiki link while writing, *should*) is planned and has no test.
- **Open technical debt:** five entries, each with its cause, consequence, fix and the event that makes it urgent (`docs/tech-debt.md`).
- **Gaps in the chain** from requirement to code to test: none.

Entries are closed only on evidence. `DEBT-026` was found only by running the stand on old volumes, and closed with the one-off fix written into the install guide. `DEBT-023`, a browser test that failed in 2 of 5 full runs and never alone, stayed open after a first fix that nothing proved; it was closed on 10 October, when the field's state recorded after a failure showed the cause — the tag list reopened by a focus handler in Tom Select, which a fast typist could hit too — and one setting took the failures to 0 in 10 runs.

4 We can explain any part of the code

How the code was written. Every prompt, the outcome and the human's own edits are in `docs/ai/journal/`, committed by a hook. Before any change the assistant states a contract (goal, boundaries, criteria, steps) and waits for confirmation; requirements, ADRs, the route contract and the schema change only with the human's approval. A failing test comes before the code, and a decision against a library is written down with the reason.

What checks it.

- 638 application tests and 356 for the tools, on every push (`scripts/check.sh`); the same suite on PostgreSQL 17 in a CI job of its own.
- 151 browser tests in Chromium through Playwright, with axe for accessibility, at four widths from 320 to 1920 pixels.
- Mutation testing (PIT, by hand before a stage): 132 mutants of the pure logic, 130 killed; the two survivors are equivalent and explained (`docs/verification/mutation-testing.md`).
- Property-based tests (jqwik) of the address rules and the wiki-link reading, 1000 generated titles each in English, Hindi and Tamil.

A map of the code. `docs/viva/README.md` gives each of the thirteen slices its routes, the class to start reading at and its tests; three request flows file by file; and where a typical change

goes — a limit, a form rule, a column, a route — with the test that checks it. We both prepare on every slice, since we take tasks freely rather than owning fixed areas.

5 Since the design document, and what comes next

The design document of 11 September planned phase 3 as the main flow. It is built: 22 features, 22 ADRs, every slice of the module map now holds code, and `ModularityTest` still enforces their boundaries. Much of phase 4 was done alongside: the handoff package that lets OGE run the guide without us (`docs/handoff/`: installing it behind IITM's HTTPS proxy, a moderator's guide with screenshots, backups restored and checked), and the limit on password guessing made to count each visitor apart behind the proxy, with tests from a trusted and an untrusted address. Ahead: the meeting with OGE on the working stand, the stakeholder using it himself, and the open debt, each at its trigger.